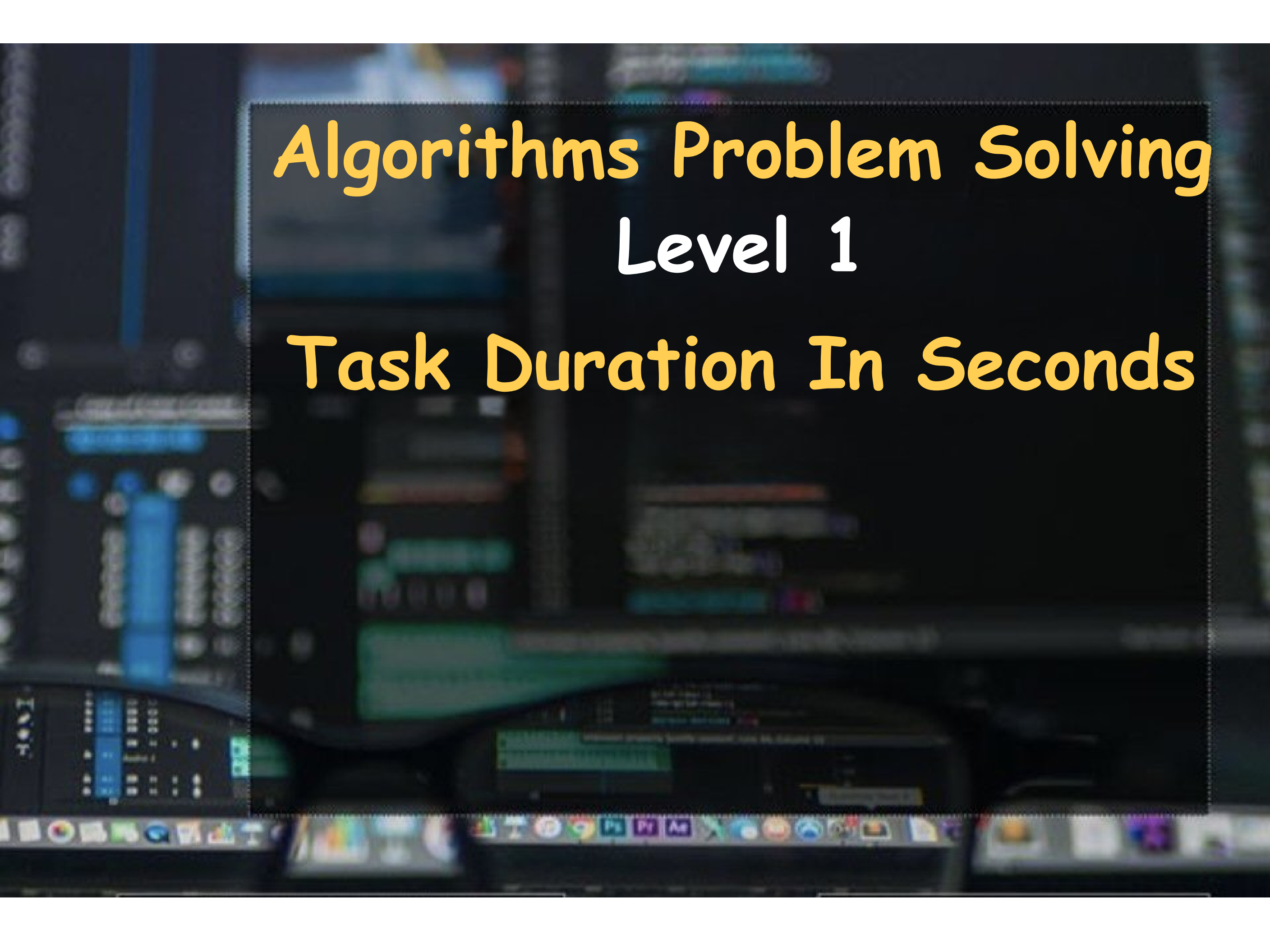
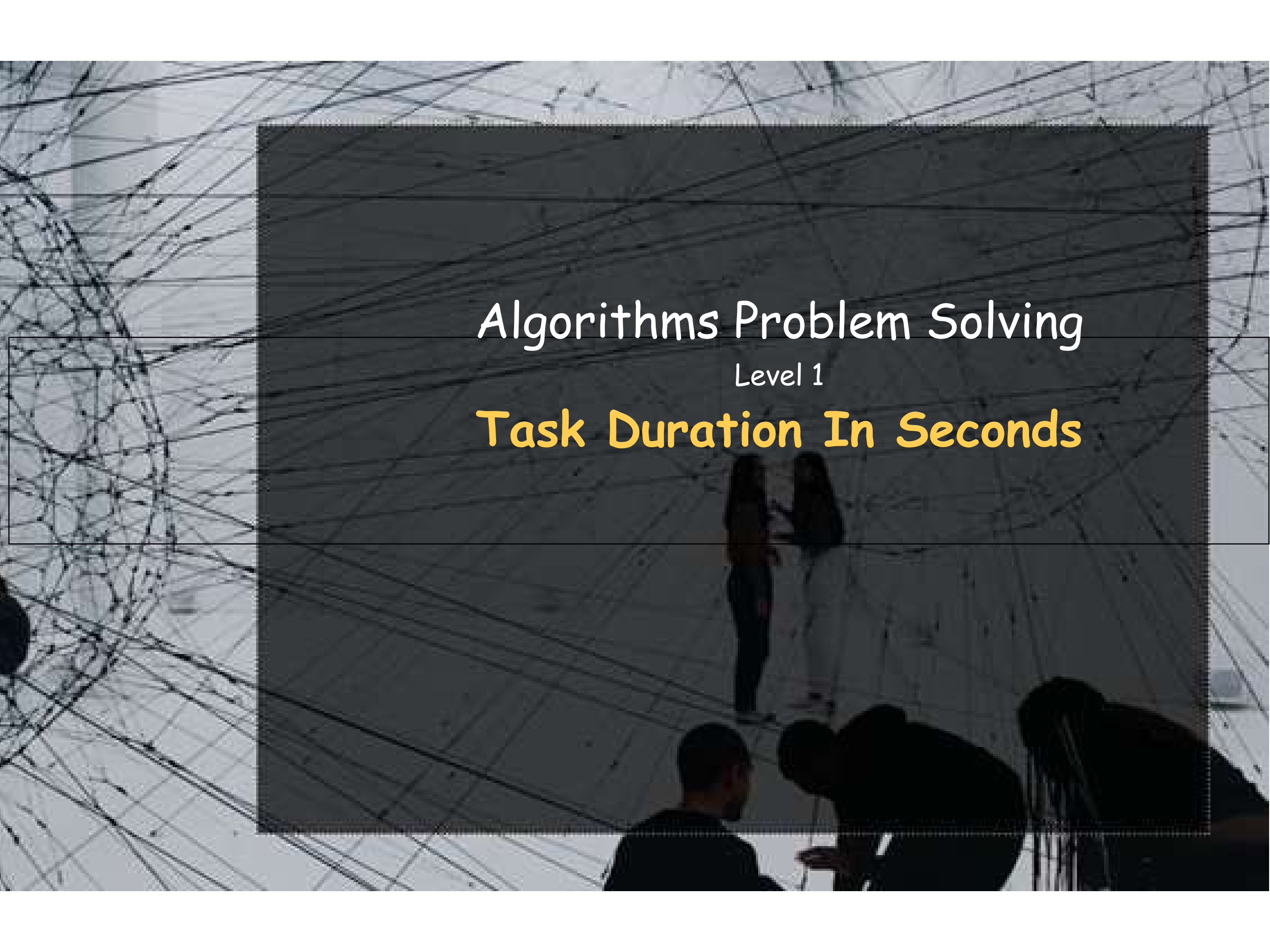




Algorithms Problem Solving Level 1

Task Duration In Seconds



Algorithms Problem Solving

Level 1

Task Duration In Seconds

Definition

- ◆ This program reads a task duration in days, hours, minutes, and seconds, then converts the entire duration into seconds.
- ◆ The program uses a structure to store the four time values.
- ◆ The total duration is calculated by converting each unit into seconds and then adding them together.

Explanation

◆ `strTaskDuration` stores:

NumberOfDays

NumberOfHours

NumberOfMinutes

NumberOfSeconds

◆ ReadTaskDuration() asks the user to enter each part of the task duration and returns the complete structure.

◆ TaskTotalSecond() converts each time unit into seconds:

1 day = $24 \times 60 \times 60$ seconds

1 hour = 60×60 seconds

1 minute = 60 seconds

1 second = 1 second

◆ All converted values are added together to get the total number of seconds.

◆ Printresult() displays the calculated total.

◆ In main(), the structure returned by ReadTaskDuration() is passed directly to Printresult().

Steps in main()

1. Read the number of days.
2. Read the number of hours.
3. Read the number of minutes.
4. Read the number of seconds.
5. Convert days to seconds.
6. Convert hours to seconds.
7. Convert minutes to seconds.
8. Add all values together.
9. Print the total duration in seconds.

💡 Example

◆ If the user enters:

Days: 2

Hours: 5

Minutes: 45

Seconds: 35

◆ The calculation is:

$$2 \times 24 \times 60 \times 60 = 172800$$

$$5 \times 60 \times 60 = 18000$$

$$45 \times 60 = 2700$$

$$35 = 35$$

◆ Total:

$$172800 + 18000 + 2700 + 35 = 193535$$

➡ Result

Total Task duration in seconds = 193535 Seconds

Note

- ◆ The program converts every time unit into the smallest unit, seconds, before adding them together.
- ◆ This is a common technique when working with different units of time.

⊘ Warning

◆ The program does not validate the input ranges. For example, it accepts 70 minutes even though a normal minute value would usually be between 0 and 59.



Comparison

▨ Converting everything to seconds

- ✓ Days, hours, minutes, and seconds are converted into one common unit.
- ✓ The values can then be added directly.
- ✓ The result is one total number of seconds.

▤ Keeping different time units

- ✓ Days, hours, minutes, and seconds remain separate.
- ✓ Direct addition between different units is not meaningful.
- ✓ Conversion is required before calculating a total.

✦✦ Summary

- ✓ A structure can group multiple related time values.
- ✓ Different time units can be converted into one common unit before calculation.
- ✓ 1 day = 86,400 seconds.
- ✓ 1 hour = 3,600 seconds.
- ✓ 1 minute = 60 seconds.
- ✓ The final result is calculated by adding all converted values together.

Thank you 😊